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Surgical stabilization of rib fractures combined with intercostal nerve cryoablation proves to be more cost effective by reducing hospital length of stay and narcotics

Zachary M. Bauman [a 1], John Loftus [a], Ashley Raposo-Hadley [a], Shradha Samuel [a], Weston Ernst [a], Charity H. Evans [a], Samuel Cemaj [a], Adam J. Kaye [b]

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Highlights

- Surgical stabilization of rib fractures (SSRF) plus intercostal nerve cryoablation (INCA) decreases postoperative narcotic requirements.
- SSRF plus INCA decreases postoperative hospital length of stay.
- Postoperative hospital charges are less for patients undergoing SSRF plus INCA.
- Total hospital charges are not statistically different for patients having SSRF plus INCA versus SSRF alone.
- SSRF plus INCA is more cost effective than SSRF alone given improved patient outcomes.

Abstract

Background

Intercostal nerve cryoablation (INCA) coupled with surgical stabilization of rib fractures (SSRF) has been shown to reduce post-operative pain scores but at what monetary cost. We hypothesize that in-hospital outcomes improve with the addition of INCA to SSRF and potential increased hospital charges are justified by patient benefits.

Methods

Multi-institutional, retrospective review of patients undergoing SSRF with and without INCA over an 8-year period. Institutions involved were Level II or higher trauma centers. Basic demographics were obtained. Patients were included if SSRF was performed during the index hospitalization. Primary outcomes included total hospital length of stay (HLOS) and HLOS after SSRF, total hospital charges (HC), HC the day of surgery and HC after surgery. Secondary outcome included total narcotic consumption in morphine milliequivalents (MME) after SSRF. Mann-Whitney U test was used for analysis. Statistical significance $p<0.05$.

Results

136 patients analyzed; 92 underwent SSRF only and 44 underwent SSRF with INCA. Demographics were similar between groups. Number of ribs stabilized was comparable; 4.78±1.64 SSRF only and 4.73±1.66 SSRF with INCA ($p=0.463$). Median ISS [16 (IQR 11.5-16) SSRF only and 14 (IQR 9-18.75) SSRF with INCA ($p=0.463$)] was not statistically different. The INCA group showed a decrease in the median total HLOS, 9 versus 10 days ($U=1517.5$, $p=0.026$) and HLOS after SSRF, 4 versus 6 days ($U=1217.5$, $p<0.001$). HC the day of surgery were higher for the INCA group, $93,932 versus $71,143 ($U=1106$, $p<0.001$). However, total HC were similar between groups and total HC after SSRF was significantly less for the INCA group, $10,556 versus $20,269 ($U=1327$, $p=0.001$). Total median narcotic use after SSRF was significantly less for the INCA group, 88.6 vs 113.7 MME ($U=1544.5$, $p=0.026$).

Conclusion

SSRF with INCA is safe and does not increase overall HC with the added benefit of decreased HLOS post-operatively and decreased narcotic consumption.

Introduction

Rib fractures affect approximately 10% of all injured patients and a staggering 55% of all blunt injuries [1,2]. Furthermore, they are associated with significant pain [3,4]. In the last decade, a major shift in the treatment of rib fractures has occurred. Although 90% of rib fractures are still treated non-operatively [5], there are a growing number of patients who will undergo surgical stabilization of rib fractures (SSRF). The indications for SSRF have changed as well. An unstable “flail” chest is no longer the only criteria for SSRF as recent studies now demonstrate those patients without a flail chest may also benefit from SSRF [4,6]. This shift in management to a more aggressive operative approach has resulted in higher charges for the patient, however, this is outweighed by the benefits of SSRF [7,8]. These benefits include decreased hospital length of stay (HLOS), decreased intensive care unit (ICU) LOS, deceased ventilator days, and decreased incidence of pneumonia [9], [10], [11]. SSRF has also resulted in earlier return to work [12] and decreased

need for narcotic pain medications [13]. Even with these improvements, patients still report a great deal of early postoperative pain during their convalescence.

In an effort to expand on the benefits of SSRF and hopefully decrease patient's pain further, intercostal nerve cryoablation (INCA) has been introduced as another adjunct in pain management. INCA is a therapy described in the 1970s that relies on the freezing of peripheral nerves in order to produce a prolonged period of analgesia [14,15]. The technique is based on the Joule-Thompson effect of rapid gas expansion which absorbs heat from the surrounding environment as the gas expands [16]. The current INCA system utilizes nitrous oxide whose properties enable the cryoprobe to reach a temperature of -60°Celsius. A peripheral nerve exposed to this temperature will experience Sunderland Stage II nerve injury, also known as axonotmesis [17]. The axon and myelin sheath are destroyed leaving the endoneural, perineural, and epineural structures intact. A recent case report demonstrated a patient treated with SSRF plus INCA had decreased post-operative pain [18]. Furthermore, another study showed that INCA, as a part of a multimodal pain regimen, had significant benefits in decreasing postoperative pain [16,19].

The addition of yet another procedure, the increased operative time, and the cost of cryoprobe has come under scrutiny. Some have questioned if the increased costs of these procedures are justified in a patient population that only a decade ago would have been treated non-operatively. The increased costs of SSRF alone have been justified [7]. However, we now need to know if this extra procedure is beneficial enough to offset potential added costs. We hypothesized that in-hospital outcomes are better with the addition of INCA to SSRF as opposed to SSRF alone, suggesting SSRF plus INCA is more cost effective then SSRF alone. The purpose of this study was to compare outcomes and hospital charges for patients undergoing a combination of SSRF and INCA to a historical group of patients treated with SSRF alone.

Section snippets

Methods

This is an Institutional Review Board approved, multi-institutional, retrospective review of all patients undergoing SSRF with and without INCA over an 8-year period. All institutions involved in the study were Level II or higher designated trauma centers and located in urban settings. Patients were included in the review if they were 18-years or older and underwent SSRF during their index hospital admission. All patients experienced a blunt traumatic mechanism of injury. All delayed SSRF …

Results

A total of 136 patients were analyzed; 92 (67.6%) who underwent SSRF only and 44 (32.2%) who underwent SSRF plus INCA. Basic demographics between the two cohorts were similar, demonstrating no statistical difference (Table 1). Of note, median ISS was not statistically different for the two groups; 16 (Inter-quartile range (IQR) 11.5 - 16) for SSRF only and 14 (IQR 9 - 18.75) for SSRF with INCA (p= 0.463). Furthermore, there was no significant differences between median c-AIS for SSRF only …

Discussion

To the authors knowledge, this is the first study examining the cost effectiveness of SSRF plus INCA. In this retrospective study of 136 patients with rib fractures, we demonstrated that SSRF plus INCA, despite resulting in more charges to the patient the day of surgery, is actually more cost effective for the patient overall. By showing the total hospital charges are the same for both cohorts and the postoperative charges are less for the SSRF patient undergoing INCA, this further supports our …

Conclusion

This is the first study to examine the cost effectiveness of INCA coupled with SSRF. Although there is an increase in hospital charges the day of surgery, overall hospital charges for the SSRF only cohort versus the SSRF plus INCA cohort are not statistically different. Furthermore, HLOS and opioid use after surgery is significantly reduced when performing SSRF in conjunction with INCA. Given the results in this study, SSRF plus INCA is more cost effective than SSRF alone given improved patient …

Declaration of Competing Interest

Dr. Bauman is a paid educating faculty for Zimmer-Biomet and KLS-Martin. Dr. Kaye is a paid educating faculty for Zimmer-Biomet and Atricure. All other authors have no additional conflicts of interest. All authors contributed substantially to this research project. All authors were fully involved in this research project and collectively designed, conducted and interpreted the data. Furthermore, all authors reviewed and approved the decision to submit this manuscript for publication in its …

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References (32)

Z.M. Bauman *et al.*
Taking the bull by the horns: patient trampled by bull requiring surgical fixation of multiple rib fractures including rib 11
Trauma Case Reports (2018)

S. Majercik *et al.*
Long-term patient outcomes after surgical stabilization of rib fractures
Am J Surg (2014)

K.M. Nelson *et al.*
Intraoperative intercostal nerve freezing to prevent postthoracotomy pain
Ann Thorac Surg (1974)

J.W. Lloyd *et al.*

A new approach to pain relief
Lancet (1976)

F.M. Pieracci *et al.*
Consensus statement: surgical stabilization of rib fractures rib fracture colloquium clinical practice guidelines
Injury (2017)

F.C. Detterbeck
Efficacy of methods of intercostal nerve blockade for pain relief after thoracotomy
Ann Thorac Surg (2005)

C.M. Harbaugh *et al.*
Comparing outcomes with thoracic epidural and intercostal nerve cryoablation after Nuss procedure
J Surg Res (2018)

B.A. Keller *et al.*
Intercostal nerve cryoablation versus thoracic epidural catheters for postoperative analgesia following pectus excavatum repair: preliminary outcomes in twenty-six cryoablation patients
J Pediatr Surg (2016)

C. Graves *et al.*
Intraoperative cryoanalgesia for managing pain after the Nuss procedure
J Pediatr Surg (2017)

L.C. Müller *et al.*
Intraoperative cryoanalgesia for postthoracotomy pain relief
Ann Thoracic Surg (1989)

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Intercostal nerve cryoablation during surgical stabilization of rib fractures decreases post-operative opioid use, ventilation days, and intensive care days
2023, Injury

> *Citation Excerpt :*
> …Our study, consistent with previous ones, demonstrates no additional time in the OR when cryoablation is used. Additionally, previous studies demonstrate significant cost savings by decreasing hospital length of stay both in the trauma population and the pectus population [11,16,31,32]. A recent randomized control trial reinforces this and shows superiority of cryoablation during the Nuss procedure over intrathoracic epidural [32].…

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2021, Journal of Thoracic Disease

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1 This research was presented via podium presentation during the 2020 Chest Wall Injury Society Virtual, April 23-24 in Denver, CO.

Dr. Zachary Bauman is a paid educating faculty for Zimmer-Biomet and KLS-martin. Dr. Adam Kaye is a paid educating faculty for Zimmer-Biomet and AtriCure. Otherwise, no other authors have any conflicts of interest.

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